

Notes: Berger, Miller, Petersen, Rajan, and Stein (JFE, 2005)

Does Function Follow Organizational Form? Evidence from the Lending Practices of Large and Small Banks

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1 Big picture

- **Question.** Does organizational form affect what a financial intermediary can do well? In particular, are small banks better than large banks at small-business lending when lending requires “soft” information?
- **Setting.** The paper studies U.S. small-business lending using the 1993 National Survey of Small Business Finance, matched to bank Call Reports and branch-location data.
- **Core theory.** The paper builds on Stein’s model of decentralized versus hierarchical firms. Small organizations keep decision authority close to information collection, which gives agents stronger incentives to collect soft information. Large organizations are better when information can be hardened, documented, and transmitted up a hierarchy.
- **Application.** Small-business lending is a natural test because many small firms have limited accounting records, limited public histories, and need lenders to learn through local contact and relationships.
- **Main empirical idea.** If small banks have a comparative advantage in soft-information lending, then firms that are more informationally opaque should tend to borrow from small banks, and small-bank relationships should look more local, personal, long-lived, exclusive, and credit-relieving.
- **Endogeneity challenge.** Firms choose their banks. Difficult borrowers may intentionally select small banks, so simply regressing relationship outcomes on bank size can mix causal effects with matching.
- **Identification strategy.** The paper instruments for the size of the bank a firm borrows from using local market banking structure and state branching regulation: $\log(\text{Median Bank Size})$ in the firm’s market and an Open Market branching-law variable.
- **Main results.** Large banks lend to firms with more hard information, such as larger firms and firms with financial records. After instrumenting for bank size, large banks lend at greater distances, use more impersonal communication, have shorter and less exclusive relationships, and do less to relieve credit constraints.
- **Economic takeaway.** Organizational form matters for function. Small banks are not just smaller versions of large banks; they appear to specialize in relationship lending based on soft information. Bank consolidation can therefore affect credit access for informationally opaque small firms.

2 Data and measurement (key objects)

2.1 Data sources and sample

- **Primary firm data.** The main source is the Federal Reserve’s 1993 National Survey of Small Business Finance (NSSBF).
- **Survey timing.** The survey was conducted in 1994–1995 and covers firms in existence at the end of 1993. Some loan information, including the most recent loan, refers to 1994.
- **Firm eligibility.** Firms must be for-profit entities with fewer than 500 employees.
- **Small-firm nature of the sample.** The mean firm has book assets of about \$3.0 million; the median firm has assets of about \$680,000.
- **Loan focus.** The paper focuses on the firm’s most recent bank loan. Each observation is a firm whose most recent loan came from a bank and was originated between 1990 and 1994; 88% of the loans were originated in 1993 or 1994.
- **Bank data.** The authors match each firm to its bank using bank Call Reports for balance-sheet variables and the FDIC Summary of Deposits for branch locations.
- **Baseline sample size.** The main sample contains **1,131** firms with information on the most recent lender.
- **Trade-credit subsample.** Tests using the fraction of trade credit paid late use **546** firms because trade-credit information is not available for all sample firms.

2.2 Soft information, hard information, and organizational form

- **Soft information.** Information that is costly or impossible to verify and transmit in a report. Examples include a loan officer’s judgment about an entrepreneur’s honesty, competence, persistence, or local reputation.
- **Hard information.** Information that can be documented and transmitted across a hierarchy. Examples include financial statements, balance sheets, income statements, audited records, collateral values, and credit scores.
- **Small-bank logic.** In a small bank, the person collecting information is closer to the person allocating capital. This gives the local lender confidence that effort spent learning about a borrower will be useful.
- **Large-bank logic.** In a large bank, decision authority is more separated from local information collection. Soft information may not be credibly transmitted to headquarters, so local incentives to collect it are weaker.

- **Important distinction.** Large banks are not predicted to be worse at all lending. They may actually do well when information is hard and can be passed through the organization.

2.3 Relationship and lending-method variables

The paper constructs five main variables that describe the lending relationship between a firm and its bank.

- **Distance.** Miles between the firm and the bank branch or office it uses most often.
- **Impersonal Relationship.** Dummy equal to one if the firm mainly interacts with the bank by phone or mail rather than in person.
- **Relationship Length.** Number of years the bank and firm have interacted through lending, deposits, or other services.
- **Single Lender.** Dummy equal to one if the bank making the most recent loan is the firm's only lender, including both bank and nonbank lenders.
- **Trade Credit Paid Late.** Fraction of trade credit the firm reports paying after it is due. This is interpreted as a proxy for credit constraints because late trade credit is expensive financing.

2.4 Bank, market, firm, and contract variables

- **Bank Size.** Assets of the firm's bank. This is the key explanatory variable.
- **Number of Branches in Market.** Number of branches that the firm's bank has in the firm's MSA or non-MSA rural county.
- **Bank Age.** Years the bank has existed.
- **Median Bank Size.** Median size of banks in the firm's market, weighted by branch counts. This is one instrument for bank size.
- **Open Market.** Fraction of 1983–1992 during which the firm's state was neither a unit-banking nor limited-branching state. This is the second instrument for bank size.
- **Market Herfindahl.** Concentration of the local banking market.
- **Firm controls.** Firm size, firm age, loan amount, line-of-credit dummy, collateral dummy, checking-account dummy, MSA dummy, industry dummies, and loan-year dummies.
- **Records.** Dummy equal to one if the survey respondent had documentation such as financial statements or accounting records to answer the survey questions. This is used as a proxy for the existence of hard information.

2.5 Key descriptive facts

- In the full sample, mean distance to the bank is **26.1 miles**, but the median is only **3 miles**, so distance is highly skewed.
- About **29.4%** of firms communicate impersonally with their bank.
- Average relationship length is **8.75 years**; the median is **6 years**.
- About **49.9%** of firms have a single lender.
- The mean fraction of trade credit paid late is **35.2%**.
- Bank size varies widely: the 25th percentile is \$163 million, the median is \$956 million, and the 75th percentile is \$7.685 billion.
- Across bank-size classes, firms borrowing from banks above \$10 billion have much larger loans and firms than firms borrowing from banks below \$100 million.
- Firms with financial records are more common among large-bank borrowers: **47.4%** among the smallest-bank group versus **65.4%** among the largest-bank group.
- Lending methods vary in the theory-predicted direction: distance rises from **14.9 miles** for the smallest banks to **71.4 miles** for the largest banks; impersonal communication rises from **16.8%** to **40.6%**; exclusive relationships fall from **61.6%** to **41.0%**.

2.6 Unobservables and timing

- The central unobservable is borrower opacity or credit difficulty. This affects both the bank a firm chooses and the form of the relationship.
- The paper treats the most recent loan as the key match between borrower and bank, then asks whether bank size predicts the relationship technology used in that match.
- Most loans are recent relative to the survey, which helps align firm characteristics, bank characteristics, and lending relationships.
- The paper does not observe a randomized assignment of firms to banks. Its empirical design therefore relies on instruments that shift the local availability of small versus large banks.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual mechanism: soft versus hard information

Let a loan officer choose effort e to produce information about a borrower. The payoff to effort depends on whether the information can be used in the final capital-allocation decision.

- If information is **soft**, it cannot be credibly transmitted to distant superiors. In a hierarchy, the local officer may collect information but later receive no capital allocation, so the private return to effort is low.
- If information is **hard**, it can be summarized in reports. Then a local officer inside a large bank can use hard evidence to advocate for a larger capital allocation.

A useful way to summarize the comparative advantage is:

Small banks: better for soft information, Large banks: better for hard information.

Interpretation. The theory does not say small banks are always better. It says the best organizational form depends on whether the lending technology relies on soft local judgment or hard verifiable information.

3.2 Bank choice and hard-information sorting

The first empirical question is whether firms with more hard information borrow from larger banks. The basic specification is:

$$\log(\text{BankSize}_i) = \alpha + \delta \text{Records}_i + X_i' \beta + C_i' \gamma + D_{\text{industry}} + D_{\text{year}} + u_i,$$

where:

- BankSize_i is assets of the bank making the firm's most recent loan;
- Records_i proxies for hard information about the firm;
- X_i includes firm size, firm age, MSA status, and other firm characteristics;
- C_i includes loan amount, line-of-credit status, collateral, and checking-account status.

Interpretation. A positive δ means that firms with formal documentation are more likely to borrow from larger banks, holding size and other characteristics fixed. This is the first piece of evidence that large banks specialize more in hard-information lending.

3.3 First-stage equation for the IV strategy

Because firms choose their banks, bank size is endogenous in relationship-outcome regressions. The paper instruments for $\log(\text{BankSize}_i)$ using local market bank size and branching-law permissiveness:

$$\log(\text{BankSize}_i) = \pi_0 + \pi_1 \log(\text{MedianBankSize}_{m(i)}) + \pi_2 \text{OpenMarket}_{s(i)} + Z_i' \pi_3 + v_i.$$

Here:

- $m(i)$ is the firm's local banking market, defined as the MSA or rural county;
- $s(i)$ is the firm's state;
- Z_i contains the same controls used in the relationship-outcome regressions.

Interpretation. If a firm is located in a market dominated by large banks, or a state where branching laws historically allowed larger banks to grow, it is more likely to be matched with a large bank for reasons partly external to the firm itself.

3.4 Continuous relationship outcomes: distance and relationship length

For continuous outcomes, the basic specification is:

$$Y_i = \alpha + \theta \log(\text{BankSize}_i) + B_i' \lambda + X_i' \beta + C_i' \gamma + D_{\text{industry}} + D_{\text{year}} + \varepsilon_i,$$

where Y_i is either:

$$\log(1 + \text{Distance}_i) \quad \text{or} \quad \log(1 + \text{RelationshipLength}_i).$$

The vector B_i includes bank and market controls, such as number of branches in the market, bank age, and market Herfindahl.

Predictions.

- For distance: $\theta > 0$. Large banks should lend farther away because hard-information lending is less distance-sensitive.
- For relationship length: $\theta < 0$. Small banks should have longer relationships because soft information is accumulated over time and is harder for borrowers to transfer to other lenders.

3.5 Binary relationship outcomes: impersonal interaction and exclusive lending

For binary outcomes, the paper estimates logit models. The latent index is:

$$Y_i^* = \alpha + \theta \log(\text{BankSize}_i) + B_i' \lambda + X_i' \beta + C_i' \gamma + D_{\text{industry}} + D_{\text{year}} + \varepsilon_i,$$

and:

$$\Pr(Y_i = 1) = \Lambda(Y_i^*) = \frac{\exp(Y_i^*)}{1 + \exp(Y_i^*)}.$$

The two binary outcomes are:

- $Y_i = 1$ if the relationship is impersonal, meaning phone or mail rather than in-person communication;
- $Y_i = 1$ if the firm has a single lender.

Predictions.

- For impersonal interaction: $\theta > 0$. Large banks should rely less on face-to-face soft-information collection.
- For single-lender status: $\theta < 0$. Small banks should form more exclusive relationships because soft information creates relationship-specific value.

3.6 Credit constraints: trade credit paid late

The paper uses late trade-credit payment as a proxy for credit constraints. The latent Tobit equation is:

$$TC_i^* = \alpha + \theta \log(\text{BankSize}_i) + B_i' \lambda + X_i' \beta + C_i' \gamma + D_{\text{industry}} + D_{\text{year}} + \varepsilon_i,$$

with observed outcome:

$$TC_i = \min\{1, \max\{0, TC_i^*\}\}.$$

Interpretation. If small banks are better at relationship lending, then firms forced to borrow from large banks should be more credit constrained. The prediction is $\theta > 0$: large-bank borrowers should pay more trade credit late.

3.7 Hausman-style endogeneity test

For the trade-credit specification, the paper also includes the first-stage residual:

$$TC_i^* = \alpha + \theta \log(\widehat{\text{BankSize}}_i) + \rho \widehat{v}_i + \text{controls} + \varepsilon_i.$$

Interpretation. A significant ρ indicates that bank size is endogenous. The paper finds a negative residual effect in the trade-credit model, consistent with the idea that more constrained or difficult firms endogenously seek out smaller banks.

3.8 Bank size versus holding-company size

Many sample banks belong to multi-bank holding companies. The paper tests whether outcomes depend on the size of the borrowing firm’s own bank or instead on the size of the broader holding company:

$$Y_i = \alpha + \theta_1 \log(\text{OwnBankSize}_i) + \theta_2 \log(\text{OtherHoldingCompanyAssets}_i) + \text{controls} + \varepsilon_i.$$

Interpretation. The important size margin is the size of the bank that actually lends to the firm, not the size of the rest of the holding company. This supports the idea that credible decentralization within a holding company can preserve small-bank-like incentives.

4 Identification and instruments (what makes the estimates credible)

4.1 Why OLS is not enough

- Bank-firm matching is endogenous: firms choose banks, and banks choose borrowers.
- Opaque or troubled firms may deliberately seek small banks because small banks are better at evaluating soft information.
- Large firms with good records may naturally select large banks because their information can be processed in standardized ways.
- Therefore, OLS estimates of how bank size affects distance, communication, relationship strength, or credit constraints can be biased.

4.2 The two instruments

The paper uses two instruments for $\log(\text{Bank Size})$:

- **Local median bank size.** The log of the median bank size in the firm’s MSA or rural county, weighted by the number of branches. This captures how available large versus small banks are

in the local market.

- **Open Market.** The fraction of the previous ten years during which the firm’s state allowed relatively unrestricted branching. More permissive branching regulation makes it easier for large banks to expand and dominate local markets.

First-stage strength. In the first-stage regression, both instruments strongly predict the size of the bank from which the firm borrows. The coefficient on $\log(\text{Median Bank Size})$ is 0.222 with a large t-statistic, and the coefficient on Open Market is 0.438.

4.3 Exclusion restriction and interpretation

- The identification assumption is that, conditional on firm, loan, bank, market, industry, and year controls, local bank-size availability and state branching history affect relationship outcomes through the size of the bank the firm is pushed toward.
- Intuition: if a small firm borrows from a large bank because no small banks are nearby, that match is less likely to reflect the firm’s own unobserved preferences or opacity.
- The IV coefficient is therefore interpreted as the effect of being forced toward a larger bank by local banking structure.
- This is why the IV estimates are often larger in magnitude than OLS: difficult borrowers may self-select into small banks, which biases OLS estimates toward zero.

4.4 Controls that help credibility

- **Firm and loan controls** absorb observable borrower size, age, loan size, loan type, collateral, and checking-account ties.
- **Records** helps separate hard-information availability from bank size.
- **Bank controls** include bank age and the number of branches the bank has in the firm’s local market.
- **Market controls** include market Herfindahl and MSA status.
- **Industry and loan-year dummies** absorb broad sectoral and time differences.
- The paper also reports robustness checks adding controls for bank mergers, bank health, and firm risk, with no material change in the main conclusions.

4.5 Limitations

- The instruments are not randomized experiments, so the exclusion restriction is ultimately an economic assumption.

- The data are cross-sectional around the early 1990s, so the paper cannot fully trace dynamic adjustment after consolidation.
- The “Records” variable is a proxy for hard information, not a direct audit of the firm’s information environment.
- The trade-credit test has a smaller sample, which reduces precision.
- Some outcomes, such as communication mode, may be mechanically related to distance. The paper discusses this and notes that distance absorbs part of the communication result.

4.6 Relation to the literature

- **Theory of organizations.** The paper connects empirical banking evidence to Coase-style questions about firm boundaries and to Stein’s theory of information production and capital allocation in hierarchies.
- **Relationship lending.** It complements work showing that bank relationships reduce credit constraints, but asks which organizational forms are best at producing such relationships.
- **Bank consolidation.** Prior work shows large banks hold fewer small-business loans or reduce small-business lending after mergers. This paper adds mechanism: large banks are less suited to soft-information lending.
- **Small-bank specialness.** The result is not simply that small banks are nice or local; it is that their organizational structure gives them a comparative advantage in soft-information-intensive tasks.

5 Tables and figures

5.1 Table 1: summary statistics and patterns by bank size

Read:

- Panel A shows the basic sample: 1,131 firms, mean distance of 26.1 miles, mean relationship length of 8.75 years, and about half of firms with a single lender.
- Panel B shows the raw bank-size gradient. Firms borrowing from the smallest banks are smaller, have smaller loans, and are less likely to have financial records.
- The raw relationship patterns already point in the theory’s direction: large-bank borrowers are farther from their banks, communicate more impersonally, and have less exclusive relationships.
- The table is descriptive, not causal. It motivates the regression and IV analysis because firm and loan characteristics also vary strongly across bank-size categories.

Table 1

Summary statistics Panel A: full sample

Panel A contains summary statistics for the variables used in all subsequent estimation. Distance is the distance between a firm and the bank branch or office it uses most often. Impersonal Relationship equals one if the firm interacts with its bank most often by phone or mail and zero if the interaction is in person. Relationship Length is the number of years the bank and the firm have been interacting (through lending, deposit, or service activities). Single Lender is a dummy variable that equals one if the firm has a single lender. Trade Credit Paid Late is the fraction of its trade credit the firm reports paying when it is past due. Bank Size is the assets of the bank from which the firm has its most recent loan. Number of Branches in Market is the number of branches which the *firm's bank* has in its market (MSA or county). Median Bank Size is the size of the median bank in the *firm's market* (MSA or county) weighted by branches. Open Market is the fraction of the previous ten years during which there were no restrictions on within-state branching in the firm's state. Firm Size is the assets of the firm. Loan Amount is the size of the most recent loan. Records is a dummy variable that equals one if the person answering the income statement and balance sheet questions for the firm had documentation such as financial statements or accounting records to help answer the questions. There are 1,131 observations in the sample.

Variable	Mean	Std Dev	25%	50%	75%
<i>Lending methods</i>					
Distance (miles)	26.053	136.992	1.000	3.000	10.000
Impersonal relation (1 = yes)	0.294	0.456	0.000	0.000	1.000
Relationship length (yrs)	8.750	7.508	3.000	6.000	12.000
Single lender (1 = yes)	0.499	0.500	0.000	0.000	1.000
Trade credit paid late	0.352	0.208	0.250	0.250	0.500
<i>Bank characteristics</i>					
Bank size (\$B)	8.883	23.147	0.163	0.956	7.685
No. of branches in market	21.486	45.494	1.000	5.000	25.000
Bank age (years)	75.263	43.914	39.000	80.000	106.000
Median bank size (\$b)	6.159	13.426	0.196	1.203	6.077
Open market	0.446	0.266	0.000	0.400	0.800
<i>Firm characteristics</i>					
Firm age (years)	14.842	8.865	8.000	13.000	22.000
Firm size (\$m)	3.003	7.136	0.150	0.680	2.850

Table 1 (continued)

Variable	Mean	Std Dev	25%	50%	75%
Loan amount (\$m)	1.001	3.750	0.030	0.125	0.600
Records (1 = yes)	0.570	0.495	0.000	1.000	1.000

Panel B: means by bank size

Panel B contains the means of selected variables across four categories of bank size (less than \$100M, \$100M-1B, \$1B-10B, and over \$10B in assets). Regressions estimating how the lending method variables depend upon bank size as well as on other firm and bank characteristics are contained in later tables.

Variable	<100M	100M-1B	1B-10B	10B +
<i>Lending methods</i>				
Distance (miles)	14.947	9.488	19.302	71.363
Impersonal relation (1 = yes)	0.168	0.216	0.375	0.406
Relationship length (yrs)	9.384	9.261	8.762	7.389
Single lender (1 = yes)	0.616	0.496	0.497	0.410
Trade credit paid late	0.325	0.374	0.340	0.349
<i>Bank characteristics</i>				
Bank size (\$b)	0.058	0.386	4.346	36.167
No. of branches in market	1.442	5.158	24.140	60.487
Bank age (years)	49.111	67.858	86.543	92.679
Median bank size (\$b)	2.765	4.401	5.304	12.964
Open market	0.305	0.413	0.497	0.544
<i>Firm characteristics</i>				
Firm age (years)	13.763	15.037	15.595	14.346
Firm size (\$m)	0.704	1.752	3.860	5.695
Loan amount (\$m)	0.180	0.375	1.198	2.402
Records (1 = yes)	0.474	0.562	0.576	0.654
Number of observations	190	379	328	234

5.2 Table 2: determinants of bank size

Read:

- The dependent variable is $\log(\text{Bank Size})$.
- Firms with financial records borrow from larger banks. In Column 1, the Records coefficient is 0.240 and significant at the 5% level.
- Larger firms and larger loans are also matched with larger banks.
- Column 2 is the first stage for IV. $\log(\text{Median Bank Size})$ and Open Market strongly predict bank size.
- Interpretation: hard-information firms are more naturally served by large banks, while the instruments supply variation in bank size induced by local market structure and branching regulation.

5.3 Table 3: distance between the firm and its bank

Read:

- The dependent variable is $\log(1 + \text{Distance})$.
- The OLS coefficient on $\log(\text{Bank Size})$ is 0.184; the IV coefficient is 0.296. Both are significant at the 1% level.
- The IV estimate implies that moving bank size from the 25th percentile to the 75th percentile raises predicted distance by about **114%**.

Table 2

Determinants of bank size

The dependent variable is $\text{Ln}(\text{Bank Size})$. Bank Size is expressed in thousands of dollars and Firm Size and Loan Size are expressed in dollars before taking logs. $\text{Ln}(\text{Median Bank Size})$ is the log of the median bank assets in the *firm's market* (MSA or county) weighted by branches. Open Market is the fraction of the previous ten years during which there were no restrictions on within-state branching in the firm's state. We use $\text{Ln}(\text{Median Bank Size})$ and Open Market to instrument for $\text{Ln}(\text{Bank Size})$ in the models that follow. Each regression contains dummy variables for whether the loan is a line of credit, whether it is collateralized, and whether the firm has a checking account from the bank. Records is a dummy variable that equals one if the person answering the income statement and balance sheet questions for the firm had documentation such as financial statements or accounting records to help answer the questions. Each regression also includes dummies for the firm's industry (construction, retail, or services) and the year in which the loan was secured (1992–1994). The number of observations is 1,131. Significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Independent variables	Models	
	1: OLS	2: OLS
<i>Bank and market characteristics</i>		
$\text{Ln}(\text{median bank size})$		0.222*** (0.032)
Open market		0.438*** (0.145)
$\text{Ln}(1 + \text{No. of branches})$		0.594*** (0.039)
Market Herfindahl		-0.226 (0.541)
$\text{Ln}(1 + \text{bank age})$		0.523*** (0.051)
<i>Firm and contract characteristics</i>		
$\text{Ln}(\text{firm size})$	0.125** (0.051)	0.099** (0.040)
$\text{Ln}(1 + \text{firm age})$	-0.172 (0.094)	-0.224*** (0.073)
$\text{Ln}(\text{loan amount})$	0.277*** (0.051)	0.198*** (0.040)
Line of credit (1 = yes)	0.197 (0.141)	0.110 (0.110)
Loan collateralized (1 = yes)	-0.313** (0.134)	-0.059 (0.105)
Checking account (1 = yes)	-0.467** (0.210)	-0.869*** (0.166)
Firm in MSA (1 = yes)	1.220*** (0.140)	0.040 (0.151)
Records (1 = yes)	0.240* (0.121)	0.178* (0.094)
Adjusted R^2	0.218	0.526

Table 3

Distance between the firm and its bank

The dependent variable is the log of one plus the distance (in miles) between the firm and the bank branch or office that it uses most often. $\text{Ln}(\text{Bank Size})$ is the log of bank assets. Bank Size is expressed in thousands of dollars and Firm Size and Loan Size are expressed in dollars before taking logs. In Column 2, we report instrumental-variable estimates where the instruments for $\text{Ln}(\text{Bank Size})$ are $\text{Ln}(\text{Median Bank Size})$, the log of the median assets of banks in the area where the firm is located, and Open Market, the fraction of the previous ten years during which there were no restrictions on within-state branching in the firm's state. The number of branches in the market includes only branches of the bank from which the firm borrows. Each regression contains dummy variables for whether the loan is a line of credit, whether it is collateralized, and whether the firm has a checking account from the bank. Records is a dummy variable that equals one if the person answering the income statement and balance sheet questions for the firm had documentation such as financial statements or accounting records to help answer the questions. Each regression also includes dummies for the firm's industry (construction, retail, or services) and the year in which the loan was secured (1992–1994). The number of observations is 1,131. Significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Independent variables	Models	
	1: OLS	2: IV
<i>Bank and market characteristics</i>		
$\text{Ln}(\text{bank size})$	0.184*** (0.021)	0.296*** (0.078)
$\text{Ln}(1 + \text{No. of branches})$	-0.385*** (0.031)	-0.467*** (0.063)
Market Herfindahl	-0.352 (0.392)	-0.455 (0.403)
$\text{Ln}(1 + \text{bank age})$	0.002 (0.038)	-0.049 (0.052)
<i>Firm and contract characteristics</i>		
$\text{Ln}(\text{firm size})$	0.028 (0.030)	0.018 (0.031)
$\text{Ln}(1 + \text{firm age})$	-0.216*** (0.054)	-0.189*** (0.058)
$\text{Ln}(\text{loan amount})$	0.076** (0.030)	0.049 (0.035)
Line of credit (1 = yes)	0.121 (0.082)	0.107 (0.083)
Loan collateralized (1 = yes)	0.049 (0.078)	0.066 (0.079)
Checking account (1 = yes)	-0.870*** (0.125)	-0.758*** (0.147)
Firm in MSA (1 = yes)	0.255** (0.105)	0.200* (0.112)
Records (1 = yes)	-0.026 (0.070)	-0.046 (0.072)
Adjusted R^2	0.269	0.235

- The number of local branches has a negative coefficient, which makes sense mechanically: more local branches make nearby lending easier.
- Main message: large banks' lending technology is less dependent on geographic proximity, consistent with reliance on hard rather than soft local information.

5.4 Table 4: impersonal communication

Read:

- The dependent variable equals one if communication is mainly by phone or mail rather than in person.
- The logit coefficient on $\log(\text{Bank Size})$ is 0.196 in the baseline and 0.324 in the IV version.
- Based on the IV estimate, moving from the 25th to the 75th percentile of bank size raises the probability of impersonal communication from about **15%** to **38%**.
- Adding distance as a control weakens the bank-size coefficient, which is sensible because distance and communication mode are closely linked.
- Main message: large-bank lending relationships are more arm's-length and less face-to-face.

Table 4

Impersonal communication between the firm and its bank

The dependent variable is a dummy that equals one if the bank and firm communicate impersonally (by phone or mail) and zero if they communicate in person. A logit model was estimated. $\text{Ln}(\text{Bank Size})$ is the log of bank assets. Bank Size is expressed in thousands of dollars and Firm Size and Loan Size are expressed in dollars before taking logs. In Column 2, we report instrumental-variable estimates where $\text{Ln}(\text{Bank Size})$ is replaced with its predicted value based on $\text{Ln}(\text{Median Bank Size})$, the log of the median assets of banks in the area where the firm is located, and Open Market, the fraction of the previous ten years during which there were no restrictions on within-state branching in the firm's state (see Table 2, Column 2). The number of branches in the market includes only branches of the bank from which the firm borrows. Each regression contains dummy variables for whether the loan is a line of credit, whether it is collateralized, and whether the firm has a checking account from the bank. Records is a dummy variable that equals one if the person answering the income statement and balance sheet questions for the firm had documentation such as financial statements or accounting records to help answer the questions. Each regression also includes dummies for the firm's industry (construction, retail, or services) and the year in which the loan was secured (1992–1994). The number of observations is 1,131. Significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Independent variables	Models	
	1: logit	2: logit/IV
<i>Bank and market characteristics</i>		
$\text{Ln}(\text{bank size})$	0.196*** (0.046)	0.324** (0.165)
$\text{Ln}(1 + \text{No. of branches})$	-0.267*** (0.064)	-0.365*** (0.131)
Market Herfindahl	-0.808 (0.907)	-0.883 (0.916)
$\text{Ln}(1 + \text{bank age})$	-0.132 (0.082)	-0.179 (0.108)
<i>Firm and contract characteristics</i>		
$\text{Ln}(\text{firm size})$	0.259*** (0.070)	0.248*** (0.070)
$\text{Ln}(1 + \text{firm age})$	-0.329*** (0.119)	-0.290** (0.124)
$\text{Ln}(\text{loan amount})$	0.082 (0.066)	0.049 (0.075)
Line of credit (1 = yes)	0.659*** (0.191)	0.642*** (0.190)
Loan collateralized (1 = yes)	0.215 (0.170)	0.245 (0.170)
Checking account (1 = yes)	-1.128*** (0.266)	-0.982*** (0.302)
Firm in MSA (1 = yes)	0.687*** (0.245)	0.608** (0.255)
Records (1 = yes)	-0.109 (0.153)	-0.115 (0.154)
Pseudo R^2	0.179	0.168

Table 5

Relationship length between the firm and its bank

The dependent variable is log of one plus the length of the relationship between the firm and its bank (in years). $\text{Ln}(\text{Bank Size})$ is the log of bank assets. Bank Size is expressed in thousands of dollars and Firm Size and Loan Size are expressed in dollars before taking logs. In Column 2, we report instrumental-variable estimates where the instruments for $\text{Ln}(\text{Bank Size})$ are $\text{Ln}(\text{Median Bank Size})$, the log of the median assets of banks in the area where the firm is located, and Open Market, the fraction of the previous ten years during which there were no restrictions on within-state branching in the firm's state. The number of branches in the market includes only branches of the bank from which the firm borrows. Each regression contains dummy variables for whether the loan is a line of credit, whether it is collateralized, and whether the firm has a checking account from the bank. Records is a dummy variable that equals one if the person answering the income statement and balance sheet questions for the firm had documentation such as financial statements or accounting records to help answer the questions. Each regression also includes dummies for the firm's industry (construction, retail, or services) and the year in which the loan was secured (1992–1994). The number of observations is 1,131. Significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Independent variables	Models	
	1: OLS	2: IV
<i>Bank and market characteristics</i>		
$\text{Ln}(\text{bank size})$	-0.048*** (0.012)	-0.150*** (0.044)
$\text{Ln}(1 + \text{No. of branches})$	0.051*** (0.017)	0.125** (0.035)
Market Herfindahl	0.313 (0.215)	0.408* (0.225)
$\text{Ln}(1 + \text{bank age})$	0.108*** (0.021)	0.155*** (0.029)
<i>Firm and contract characteristics</i>		
$\text{Ln}(\text{firm size})$	0.022 (0.016)	0.031* (0.017)
$\text{Ln}(1 + \text{firm age})$	0.607*** (0.030)	0.582*** (0.032)
$\text{Ln}(\text{loan amount})$	-0.045*** (0.016)	-0.020 (0.020)
Line of credit (1 = yes)	0.000 (0.045)	0.012 (0.046)
Loan collateralized (1 = yes)	-0.056 (0.043)	-0.072 (0.044)
Checking account (1 = yes)	0.446*** (0.068)	0.343*** (0.082)
Firm in MSA (1 = yes)	-0.165*** (0.057)	-0.115** (0.063)
Records (1 = yes)	-0.049 (0.038)	-0.031 (0.040)
Adjusted R^2	0.366	0.348

5.5 Tables 5 and 6: relationship length and exclusivity

Read:

- Table 5 studies $\log(1 + \text{Relationship Length})$. The IV coefficient on bank size is -0.150 , much larger in magnitude than the OLS coefficient of -0.048 .
- The IV estimate implies that moving from the 25th to 75th percentile of bank size cuts predicted relationship length from about **8.8 years** to **4.5 years**.
- Table 6 studies whether the firm has a single lender. The IV logit coefficient on bank size is -0.526 , compared with -0.096 in the ordinary logit.
- The IV estimate implies that moving from the 25th to 75th percentile of bank size reduces the probability of an exclusive relationship from about **74%** to **27%**.
- Main message: small-bank relationships are more durable and exclusive, as expected when relationship-specific soft information is valuable.

5.6 Table 7: credit constraints using trade credit paid late

Read:

- The dependent variable is the fraction of trade credit paid late, estimated with a two-sided

Table 6

Exclusive relationship between the firm and its bank

The dependent variable is a dummy that equals one if the bank is the firm's only lender, and zero otherwise. A logit model is estimated. Ln(Bank Size) is the log of bank assets. Bank Size is expressed in thousands of dollars and Firm Size and Loan Size are expressed in dollars before taking logs. In Column 2, we report instrumental-variable estimates where Ln(Bank Size) is replaced with its predicted value based on Ln(Median Bank Size), the log of the median assets of banks in the area where the firm is located, and Open Market, the fraction of the previous ten years during which there were no restrictions on within-state branching in the firm's state (see Table 2, Column 2). The number of branches in the market includes only branches of the bank from which the firm borrows. Each regression contains dummy variables for whether the loan is a line of credit, whether it is collateralized, and whether the firm has a checking account from the bank. Records is a dummy variable that equals one if the person answering the income statement and balance sheet questions for the firm had documentation such as financial statements or accounting records to help answer the questions. Each regression also includes dummies for the firm's industry (construction, retail, or services) and the year in which the loan was secured (1992–1994). The number of observations is 1,131. Significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Independent variables	Models	
	1: logit	2: logit/IV
<i>Bank and market characteristics</i>		
Ln(bank size)	−0.096** (0.040)	−0.526*** (0.144)
Ln(1 + No. of branches)	0.075 (0.057)	0.388*** (0.116)
Market Herfindahl	−0.637 (0.721)	−0.242 (0.732)
Ln(1 + bank age)	0.115 (0.071)	0.308** (0.095)
<i>Firm and contract characteristics</i>		
Ln(firm size)	−0.318*** (0.057)	−0.283*** (0.058)
Ln(1 + firm age)	0.246** (0.102)	0.143 (0.107)
Ln(loan amount)	0.112** (0.056)	0.219*** (0.065)
Line of credit (1 = yes)	−0.078 (0.150)	−0.027 (0.152)
Loan collateralized (1 = yes)	−0.351*** (0.142)	−0.149*** (0.145)
Checking account (1 = yes)	0.517** (0.235)	0.087 (0.271)
Firm in MSA (1 = yes)	−0.098 (0.192)	0.110 (0.204)
Records (1 = yes)	−0.120 (0.128)	−0.044 (0.131)
Pseudo R ²	0.062	0.067

Table 7

Fraction of trade credit paid late

The dependent variable is the fraction of trade credit the firm pays late. A tobit model is estimated. Ln(Bank Size) is the log of bank assets. Bank Size is expressed in thousands of dollars and Firm Size and Loan Size are expressed in dollars before taking logs. In Column 2, we report instrumental-variable estimates where Ln(Bank Size) is replaced with its predicted value based on Ln(Median Bank Size), the log of the median assets of banks in the area where the firm is located, and Open Market, the fraction of the previous ten years during which there were no restrictions on within-state branching in the firm's state (see Table 2, Column 2). The number of branches in the market includes only branches of the bank from which the firm borrows. Each regression contains dummy variables for whether the loan is a line of credit, whether it is collateralized, and whether the firm has a checking account from the bank. Records is a dummy variable that equals one if the respondent to the survey had documentation to help answer the questions. Bank Size Residual is the residual from the first-stage bank-size regression (Table 2, Column 2) and is used to conduct a test of whether bank size is exogenous. Each regression also includes dummies for the firm's industry (construction, retail, or services) and the year in which the loan was secured (1992–1994). The number of observations is 546. Significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Independent variables	Models		
	1: tobit	2: tobit/IV	3: tobit
<i>Bank and market characteristics</i>			
Ln(bank size)	0.006 (0.006)	0.044** (0.021)	0.044** (0.021)
Ln(1 + No. of branches)	−0.017** (0.008)	−0.044*** (0.017)	−0.044*** (0.017)
Market Herfindahl	0.024 (0.112)	−0.016 (0.114)	−0.016 (0.114)
Ln(1 + bank age)	0.007 (0.010)	−0.010 (0.014)	−0.010 (0.014)
Bank size residual (Hausman test)			−0.041* (0.023)
<i>Firm and contract characteristics</i>			
Ln(firm size)	−0.002 (0.009)	−0.005 (0.009)	−0.005 (0.009)
Ln(1 + firm age)	0.022 (0.015)	0.032** (0.016)	0.032** (0.016)
Ln(loan amount)	−0.008 (0.009)	−0.017* (0.010)	−0.017* (0.010)
Line of credit (1 = yes)	0.003 (0.023)	0.000 (0.023)	0.000 (0.023)
Loan collateralized (1 = yes)	0.042* (0.022)	0.050** (0.022)	0.050** (0.022)
Checking account (1 = yes)	−0.003 (0.039)	0.036 (0.045)	0.036 (0.045)
Firm in MSA (1 = yes)	0.066** (0.030)	0.047 (0.031)	0.047 (0.031)
Records (1 = yes)	0.028 (0.020)	0.022 (0.020)	0.022 (0.020)
Log likelihood	12.425	14.090	14.180

Tobit.

- In the non-IV Tobit, the coefficient on bank size is small and insignificant.
- In the IV Tobit, the coefficient on bank size is 0.044 and significant.
- The IV estimate implies that moving from the 25th to 75th percentile of bank size raises the fraction of trade credit paid late from about **26%** to **43%**.
- This is the paper's most direct real-effect result: firms forced to borrow from large banks appear more credit constrained.
- The Hausman residual is negative and significant at the 10% level, consistent with endogenous matching: firms that are naturally more credit constrained seek smaller banks when they can.

5.7 Holding-company results and absence of figures

Read:

- The paper does not rely on figures; its evidence is organized around seven tables.
- In unreported holding-company tests, the size of the rest of the holding company has little effect once the size of the firm's own bank is controlled for.

- This suggests that decentralization matters. A large holding company may preserve small-bank-like incentives if decision authority is credibly located at the bank subsidiary level.
- This result is important because it distinguishes raw organizational size from the location of actual lending authority.

6 Short summary

- **Method.** The paper matches small firms to their bank lenders and studies how bank size predicts borrower type, distance, communication mode, relationship strength, and credit constraints.
- **Identification.** Because bank-firm matching is endogenous, the paper instruments for bank size using local median bank size and state branching-law openness.
- **Main empirical result.** Large banks serve more hard-information borrowers and have lending relationships that are farther away, more impersonal, shorter, less exclusive, and less effective at easing credit constraints.
- **Interpretation.** Small banks have a comparative advantage in collecting and acting on soft information; large banks are better suited to standardized, hard-information lending.
- **Policy implication.** Bank consolidation may reduce credit access for opaque small firms not mainly through market power, but through the disappearance of lenders organizationally suited to soft-information lending.

7 Conclusion

7.1 Core conclusion

- The paper provides empirical evidence that organizational form affects economic function.
- Small banks are better at relationship lending that relies on soft, local, nonverifiable information.
- Large banks are more likely to lend where information is hard and standardized.
- Correcting for endogenous matching strengthens the evidence: IV estimates are often larger than OLS estimates, especially for relationship length, exclusivity, and credit constraints.

7.2 Economic intuition

- Soft information is useful only if the person collecting it has authority, or can credibly influence authority.

- Small banks put information collection and decision authority closer together.
- Large banks separate local information gathering from capital allocation, which weakens incentives to collect soft information.
- When information is hard, the hierarchy problem is less severe because information can be documented, compared, and transmitted.

7.3 Takeaway

This paper gives an empirical answer to a Coasian question: what kinds of tasks are different organizational forms good at? In small-business lending, organizational form appears to shape the lending technology itself. Small banks specialize in local, relationship-based lending that depends on soft information, while large banks specialize in harder, more standardized lending. The policy lesson is that consolidation can matter even when it does not simply raise concentration: it can change the type of information used in credit allocation and thereby affect opaque firms' access to finance.